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CO₂-Resistant Nano-Reinforced Gels for Effective Gas Channeling Control in CO₂ Geological Utilization

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Abstract

CO₂ flooding technology can effectively enhance oil recovery in low-permeability reservoirs while offering significant potential for CO₂ geological sequestration. However, high-permeability zones and micro-fractures within reservoirs often lead to gas channeling, thereby reducing the efficiency of CO₂ utilization. In this study, a CO₂-tolerant crosslinked polymer gel system was developed as a potential solution to mitigate gas channeling. By incorporating nanoparticles into the gel network, the stability of the gel in CO₂ environments was significantly improved. A CO₂-resistant polymer was first synthesized using acrylamide, 2-acrylamido-2-methylpropanesulfonic acid, and dimethylaminoethyl methacrylate as functional monomers. This polymer was then mixed with nano-silica and crosslinked with hexamethylenetetramine (HMTA) and hydroquinone to construct a robust viscoelastic gel network. The optimized formulation of the CO₂-resistant nano-reinforced gel was determined to be: 0.5% CO₂-resistant polymer + 0.4% HMTA + hydroquinone + 0.2% nano-silica. Under CO₂ conditions of 90 °C and 15 MPa, the CO₂-resistant nano-reinforced gel exhibited excellent anti-aging performance, with its elastic modulus increasing from 16.04 Pa to 24.99 Pa. Core flooding laboratory experiments and field trials confirmed its gas-blocking capability. After CO₂ exposure, the residual resistance factor increased from 32.78 to 45.24, the breakthrough pressure gradient rose from 25.36 to 40.25 Pa. The gas channeling development coefficient is reduced to 0.78. The injection pressure in field applications reached 25.4 MPa, all of which significantly outperformed conventional gels. The results indicate that this CO₂-resistant nano-reinforced gel possesses excellent CO₂ resistance and gas-blocking performance, providing an effective technical solution to address gas channeling issues during CO₂ geological utilization.

Keywords: CO₂ flooding, Nano-reinforced gel, CO₂ resistance, Channeling control, CO₂ geological utilization

Introduction

The CCUS-EOR (Carbon Capture, Utilization, Storage, and Enhanced Oil Recovery) technology injects the captured CO₂ into the developed reservoirs with large reservoir space, good connectivity, and complete

caprock to displace the oil [1,2]. In this process, CO₂ can effectively extract light hydrocarbon components, promote oil expansion, and reduce the viscosity of crude oil by miscible interaction, thereby significantly improving displacement efficiency and enhancing oil recovery [3–5]. Simultaneously, injected CO₂ is safely stored underground through various mechanisms such as replacement, retention, dissolution, and mineralization, achieving carbon sequestration [6–8]. The CCUS-EOR closely links carbon sources produced by the coal chemical industry, refining, and other industries with oilfield development. It generates dual benefits of increasing oil production and reducing carbon emissions along with the realization of resource recycling [9]. However, the reservoir's heterogeneity makes it difficult to balance displacement. High permeability layers and fractures are commonly developed in almost all CO₂-developed reservoirs. During CO₂ injection, non-equilibrium displacements like finger shapes or even severe channeling are likely to occur, which not only decreases the displacement capacity but also significantly affects the CO₂ storage potential [10]. Therefore, controlling the gas flow direction and expanding the swept volume is essential for the efficient geological utilization of CO₂ [11,12].

In recent years, the technology of controlling CO₂ channeling has gradually gained attention and development. *Al-Ali et al.* found that in homogeneous reservoirs, the alternate injection of water and CO₂ can effectively reduce the CO₂ relative permeability, thereby slowing down CO₂ channeling and expanding the sweep efficiency [13]. However, in non-homogeneous reservoirs, the scuttling inhibition effect of water is significantly weakened, and water also shields the contact between CO₂ and crude oil, resulting in the unsatisfactory channeling control and oil displacement effect [14]. It is necessary to construct a more efficient channeling control system. The foam system can prevent the subsequent CO₂ channeling by plugging the pore throat, a phenomenon known as the Jamin effect. However, the foam is a thermodynamically unstable system, and the short stability period limits its application in the field of long-term blocking [15–17].

Researchers have drawn from the experience of conformance control in water flooding and use high-concentration polymer gel to decrease the permeability of dominant channels. By reducing permeability differences and plugging large-size fractures, polymer gel contributes to correcting the unbalanced flow behavior of CO₂ in the reservoir and redirecting CO₂ to displace low-permeability layers, as depicted in Fig. 1 [18–20]. Furthermore, polymer gel can also serve as a remediation method for CO₂ leakage within the reservoir [21]. The polymer gel can enter the gas channeling channel in a liquid state, and gradually crosslink to form a high-strength viscoelastic solid for plugging under the action of formation temperature [22]. *Taabbodi et al.* evaluated the effect of the gel system composed of polyacrylamide and chromium acetate to prevent CO₂ channeling. Experiments show that the residual resistance coefficient of the gel system is more than 100 at 40°C and 8.27 MPa [23]. *Sydanek et al.* proved that the chromium crosslinked gel system can be applied to a broad pH range (2–12) and has good stability in the process of CO₂ flooding [24,25]. *Li et al.* [26–28] used resorcinol and urotropine as crosslinking agents to prepare gel, and the plugging rate of the system reached 90% at 90°C. The above studies have shown that polymer gel can effectively block CO₂, but *Sun et al.* [29,30] found that long-term exposure of crosslinked polymer gel under supercritical CO₂ conditions can significantly degrade and dehydrate. Traditional polymer gels face significant challenges in long-term control of CO₂ channeling, because the decrease in the strength and volume retention of polymer gels may bring the risk of failure to CO₂ channeling control. Therefore, developing novel gel systems with better stability under CO₂ conditions has become an important research focus. Some recent studies have attempted to introduce thermal-responsive or CO₂-responsive mechanisms to enhance the hydrophilicity of gels, thereby compensating for the shrinkage caused by CO₂-induced dehydration through water absorption and swelling [31,32]. This approach can moderately improve the CO₂ tolerance of the gel. However, its effectiveness is limited because the structural strength of the gel network is important for maintaining stability. If the gel network is significantly disrupted in the acidic environment caused by CO₂, even enhanced hydrophilicity will not prevent the gel from rapidly losing

stability under external stress. Thus, only improving hydrophilicity is insufficient to fundamentally enhance the gel's resistance to CO_2 . In contrast, increasing the crosslinking density of the gel is considered a more effective strategy to significantly improve its CO_2 tolerance. Higher crosslinking density can strengthen the gel's structural strength and resistance to deformation, while simultaneously increasing water retention due to a denser network [22]. However, present methods typically obtain higher crosslinking density by increasing the polymer concentration, which dramatically raises the viscosity of the gel-forming solution, making the injection process more difficult or even impossible. In recent years, inorganic fillers such as montmorillonite, and nano-silica have been widely used to enhance the three-dimensional structural strength of gels. Among these nanomaterials, nano-silica exhibits strong surface energy and a large specific surface area, with abundant hydroxyl groups on its surface, enabling easy dispersion within polymer molecular chains. Nano-silica, as a rigid nanofiller, can serve as a "physical crosslinking point" in polymer systems. Once incorporated into the polymer network, it enhances the compactness and strength of the mesh structure, reduces porosity, supports the gel structural network through physical filling, and provides a "nano-barrier" that strengthens the retention of bound water. This improves the mechanical properties and reduces ion permeability. In addition, nano-silica materials possess strong CO_2 tolerance and can further enhance the CO_2 resistance of the gel by being incorporated into the gel network structure [33]. Therefore, utilizing subsequently injected CO_2 to in situ enhance the structural strength of the gel after its formation, combined with structural reinforcement using nano-silica materials, has emerged as a promising research direction. This method involves inducing the formation of new crosslinked networks within the gel through the injected CO_2 after gelation, thereby improving the stability and plugging performance of the gel under CO_2 displacement.

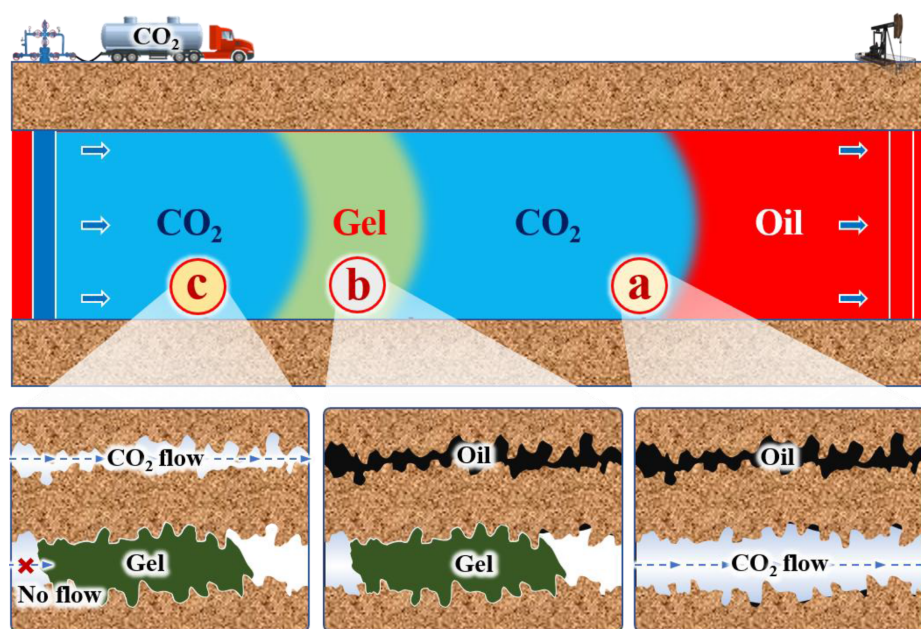


Figure 1—Schematic diagram of using gel system to control CO_2 channeling control:
 (a) CO_2 channeling control is formed, (b) Gel system forms gel barrier in the channeling control, and (c) The subsequent CO_2 flow direction is changed to increase swept volume.

This investigation first synthesized a polymer with high hydrophilicity and the ability to form supramolecular interactions between molecules after the action of CO_2 , building on previously discussed principles. Subsequently, a CO_2 -resistant nano-reinforced gel with high plugging strength and excellent CO_2 tolerance was prepared using the synthesized polymer, optimized crosslinking agent, and nano-silica materials with structural reinforcement capability. The performances and stability of this gel in

CO₂ conditions were systematically evaluated, and the mechanism of its CO₂ tolerance was clarified. This provided a promising way to solve the problem of supercritical CO₂ channeling in low permeability reservoirs.

Materials and Methods

Materials

Acrylamide (AM, 99.0%), 2-Acrylamido-2-methyl-1-propanesulfonic acid (AMPS, 98.0%), and 2,2'-Azobis(2-methylpropionamidine) dihydrochloride (V-50, 97%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. 2-(Dimethylamino)ethyl methacrylate (DMAEMA, 99%, with the stabilizer removed in advance using an alumina column), were provided by Shanghai Maclin Biochemical Technology Co., Ltd. Hexamethylenetetramine (HMTA), hydroquinone, ammonium persulfate (APS), sodium bisulfite (SBS), thiourea, and other common chemicals were supplied by Sinopharm Chemical Reagent Co., Ltd. Nano-silica was purchased from Brilliant Tengfei Nano-materials Co., Ltd. (particle size 15±5nm). The non-ionic polymer (HPAM) with a molecular weight of 800(10⁴ g/mol) was provided by Langfang Xinxing Chemical Co., Ltd. High-purity N₂ and CO₂ were obtained from Qingdao Deyi Gas Co., Ltd., and deionized water was prepared in the laboratory.

Preparation of AM/AMPS/DMAEMA copolymer (PAAD)

The synthesis process of the PAAD polymer is illustrated in Fig. 2. Initially, a specific amount of AM, AMPS, and deionized water was added to a breaker and stirred until fully dissolved. The molar ratio of AM to AMPS was maintained at 10:1. The pH of the solution was adjusted to 7.5±0.5 by adding NaOH. Subsequently, DMAEMA, equivalent to 5.5% of the total moles of monomers, was introduced and stirred until completely dissolved. The resulting mixture was transferred into a three-neck reactor and bubbled with nitrogen gas for 30 minutes to remove dissolved oxygen. Then, V-50 and a redox initiation system, along with thiourea, were added under continuous nitrogen injection. The polymerization process was conducted in two stages: the mixture was first reacted for two hours at 10°C in an ice-water bath, followed by heating to 50°C for an additional four hours. After the reaction, the resulting gel was cut into small pieces and washed three times with ethanol to remove water and residual monomers. Finally, the solid PAAD polymer was obtained through vacuum drying. The molecular weights of PAAD (470×10⁴ g/mol) were measured using a viscosimeter.

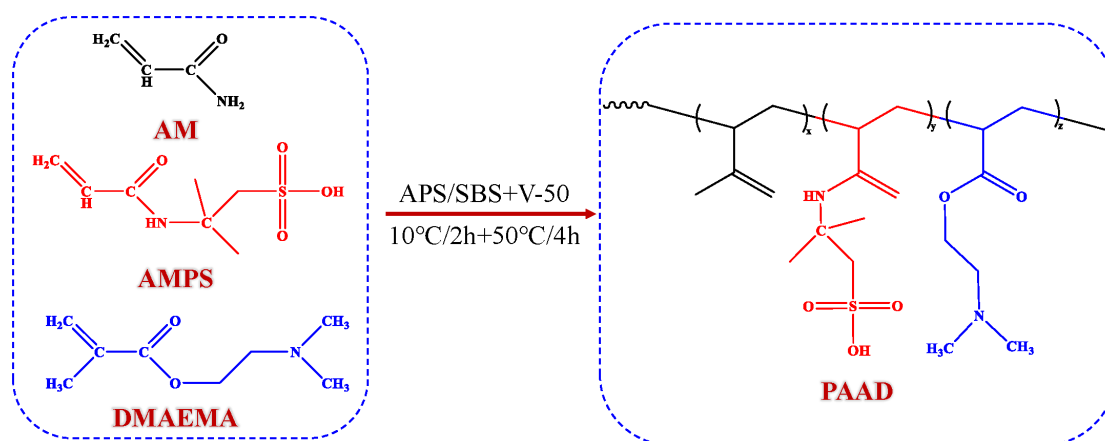


Figure 2—Graphical synthetic routes of PAAD polymer.

Characterization of PAAD polymer

The chemical structure of the synthesized PAAD polymer was characterized using a Fourier-transform infrared spectrometer (FTIR, IRTracer 100, Shimadzu, Japan) and a hydrogen nuclear magnetic resonance instrument (^1H NMR, 400 MHz, Bruker, Germany). In the FTIR experiment, the KBr pellet method was employed for sample preparation, with a scanning wavelength range of 4000 cm^{-1} – 400 cm^{-1} and a resolution of 4 cm^{-1} . Heavy water (D_2O) was used as the solvent in the ^1H NMR experiment.

Stability in a high-temperature and high-pressure CO_2 condition

To evaluate the CO_2 resistance of the synthesized polymer and the subsequent polymer gel under formation conditions, we designed a high-temperature, high-pressure CO_2 condition simulation device. The pressure vessel containing the sample to be tested was placed in a thermostat set to the specified temperature. CO_2 was then injected into the vessel using a gas compressor and gradually pressurized to the target pressure. When the temperature and pressure stabilized, the vessel was sealed. After the designated test period, the vessel was gradually cooled, and the gas pressure was released at a slow rate of 0.1 MPa/min . The sample was then taken out to observe its post-treatment condition and conducted performance testing.

Rheology measurement

A rheometer (Haake Mars 60, Thermo Fisher Scientific, Germany) was used to determine the viscosity of PAAD polymer and rheological properties of CO_2 -resistant nano-reinforced gel, including gel strength and creep behavior. Polymer viscosity measurements were performed using a high-temperature and high-pressure Hastelloy measurement system, with the shear rate range for the variable shear tests set between 0.1 s^{-1} and 1000 s^{-1} . The rheological properties of the gels were measured using a roughened plate-plate measurement system with a 1 mm gap. During the stress sweep tests, the viscoelastic modulus of the gel samples was determined at a constant frequency of 6.28 rad/s . In the creep tests, the shear stress was controlled to assess the creep-recovery behavior of the gels under constant stress, with a stress application period of 300 s followed by a recovery period of 500 s .

Core flooding and plugging experiment

The channeling control ability of CO_2 -resistant nano-reinforced gel was investigated under supercritical CO_2 flooding conditions. To ensure that injected CO_2 remained in a supercritical state during the displacement process, the back pressure was maintained at 15 MPa . The confining pressure is always 3 MPa above the injection pressure. The temperature was maintained at 90°C throughout the experiment. The procedure was as follows:

1. The size and porosity of the core were measured. The core was subsequently put in the core holder and kept in the thermostat for 12 hours to stabilize. A formation water was injected into the core at a continuous rate of 0.2 mL/min until it was completely saturated.
2. Supercritical CO_2 was injected into the core at a rate of 0.2 mL/min , and the value (P_1) after the displacement pressure difference was stabilized was recorded.
3. The gel-forming solution was injected into the core at the same flow rate, with a volume of 0.5 PV (where PV is the pore volume). The core holder was then sealed and placed in the thermostat for five days.
4. Supercritical CO_2 was injected into the core at a constant pressure of 15 MPa for some days to simulate the contact between CO_2 and the gel in the core.
5. The core was re-injected with supercritical CO_2 at a flow rate of 0.2 mL/min until the pressure stabilized. The stable displacement pressure difference (P_2) after plugging was recorded.

The residual resistance factor (F_{rr}) of the gel system was calculated using the following equations:

$$F_{RR} = P_2 / P_1 \quad (1)$$

Results and discussion

Structural analysis of PAAD polymer

Fig. 3 (a) presents the FTIR spectrum of the synthesized polymer. The absorption peaks observed at 1652 cm^{-1} and 1415 cm^{-1} can be attributed to the stretching vibration of the C=O bond in the amide group and the bending vibration of the C–N bond, respectively. Additionally, the peaks observed at 1038 cm^{-1} and 625 cm^{-1} correspond to the stretching vibrations of the S–O bond and C–S bond, while the peak at 1186 cm^{-1} indicates the presence of the asymmetric vibration of the S=O bond, suggesting the presence of the $-\text{SO}_3^-$ group in the sample [34]. The stretching vibration of the C–O bond in the ester group corresponds to the peak that appeared at 1116 cm^{-1} . These absorption peaks indicate the presence of amide, sulfonic acid, and ester groups. Fig. 3 (b) shows the ^1H NMR spectrum of PAAD. The chemical shifts of $-\text{CH}_2-$ (c) and $-\text{CH}-$ (d) on the main chain are 1.62 ppm and 2.19 ppm, respectively, while $-\text{CH}_3$ (b) and $-\text{CH}_2-$ (f) on the AMPS side chain are 1.48 ppm and 3.21 ppm, respectively. The chemical shifts of the $-\text{CH}_3$ (a) and $-\text{CH}_3$ (e) in the DMAEMA side chain are observed at 1.17 ppm and 2.91 ppm, respectively, and 3.64 ppm is the chemical shift of $-\text{CH}_2-$ (g). In addition, 4.79 ppm is the solvent peak of heavy water. Analysis of the FTIR and ^1H NMR spectra confirm the expected molecular structure of the PAAD polymer.

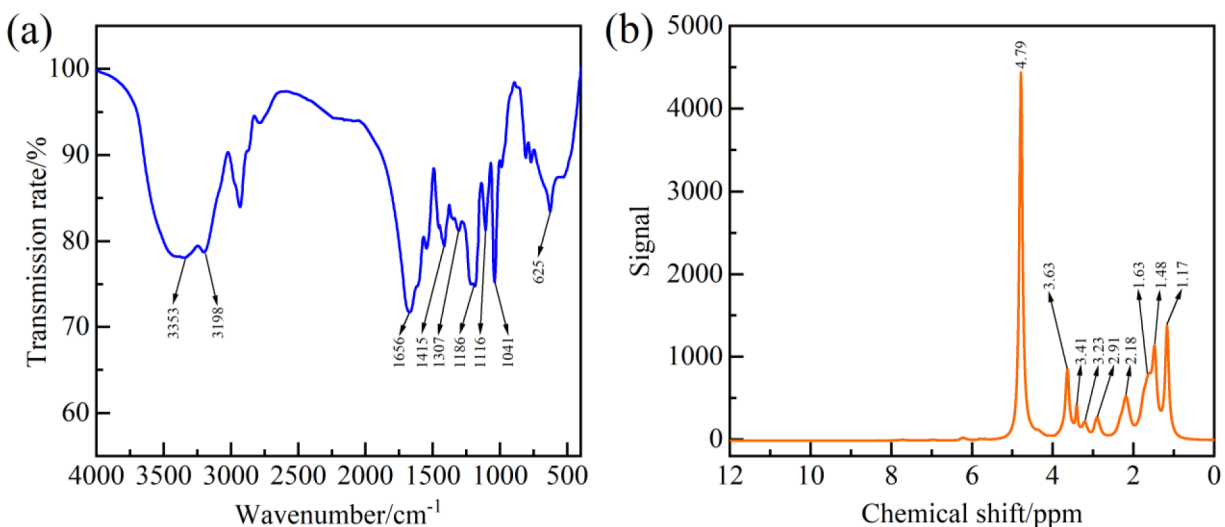


Figure 3—Structural characterization of PAAD polymer: (a) FTIR spectra; (b) ^1H NMR spectra.

Gelation performances of the CO_2 -resistant nano-reinforced gel

The gel is a viscoelastic material formed by the crosslinking of a polymer with a crosslinker, and its gelation performance is a crucial factor directly influencing the plugging effectiveness. Therefore, it is essential to investigate the strength of gel systems formed by different combinations of polymers and crosslinkers. In this study, nanosilica was first uniformly dispersed in simulated formation water. Hexamethylenetetramine (HMTA) and hydroquinone were selected as potential crosslinkers to form a high-strength gel network with the synthetic polymer. The crosslinking reaction occurred when HMTA decomposed to generate formaldehyde, which promoted the condensation and hydroxymethylation of active sites on the benzene ring of the diphenols, leading to the formation of phenolic resin. The hydroxymethyl groups ($-\text{CH}_2\text{OH}$) in the phenolic resin further crosslinked with the amide groups ($-\text{CONH}_2$) of the polymer through a dehydration reaction, transforming the viscous solution into a viscoelastic gel. As a rigid nanoscale filler, nanosilica

served as a "physical crosslinking point" within the gel network, enhancing the compactness and strength of the network structure.

The effect of the ratio between the polymer and crosslinker (It is expressed by the sum of the concentrations of HMTA and hydroquinone, in which the ratio of HMTA to hydroquinone is 1#1) on the gel strength was further investigated. As shown in Fig. 4, the elastic modulus of the gel significantly increased with the rising concentrations of both the polymer and the crosslinker. When the concentrations of the polymer and crosslinker reached 0.6 wt% and 1.0 wt%, respectively, the static elastic modulus reached 20.4 Pa, and the dynamic elastic modulus reached 23.5 Pa. This enhancement was attributed to the higher crosslinking density in the gel network and the increased entanglement and friction among flexible polymer chains. However, excessive polymer concentrations may lead to higher injection resistance and increased costs, necessitating adjustments to polymer dosage based on reservoir permeability. At a constant polymer concentration, the elastic modulus of the gel generally increased with increasing crosslinker concentration. During this process, the crosslinker strengthened the gel by forming a denser crosslinked network with the polymer. Nevertheless, as the crosslinker concentration continued to rise, the number of amide groups on the polymer chains available for crosslinking decreased, resulting in a gradual slowdown in the rate of strength improvement. At high polymer concentrations, further increases in crosslinker concentration may lead to increased gel brittleness, potentially reducing the elastic modulus.

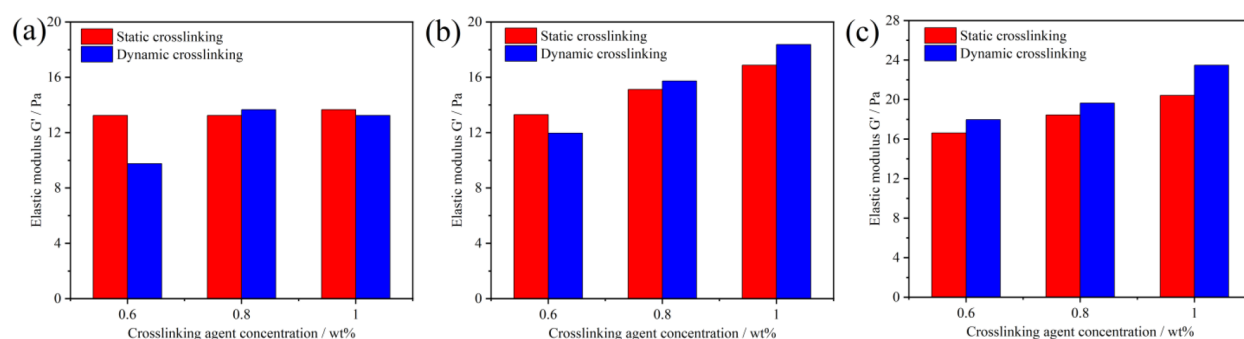


Figure 4—The effect of the ratio between polymer and crosslinker on the viscoelasticity of resulting gels: (a) 0.4 wt% PAAD polymer; (b) 0.5 wt% PAAD polymer; (c) 0.6 wt% PAAD polymer.

CO₂ resistance of the CO₂-resistant nano-reinforced gel

To evaluate the stability of gels in a CO₂ environment, the viscoelastic modulus changes of conventional HPAM gel and CO₂-resistant nano-reinforced gel systems before and after CO₂ aging were investigated. The HPAM gel system was prepared using 0.5 wt% HPAM polymer and the optimal crosslinker ratio, which includes 0.4 wt% hexamethylenetetramine and 0.4 wt% hydroquinone. The CO₂-resistant nano-reinforced gel system was prepared using 0.5 wt% PAAD polymer and the optimal crosslinker ratio, which includes 0.4 wt% hexamethylenetetramine, 0.4 wt% hydroquinone, and 0.2 wt% nano-silica. The CO₂ environment was set at a temperature of 90 °C and a pressure of 15 MPa. As shown in Fig. 5(a), the experimental results indicate that after CO₂ aging, the elastic modulus of the linear viscoelastic region of the conventional gel decreases from 19.22 Pa to 10.49 Pa, while the viscous modulus drops from 2.46 Pa to 0.78 Pa. The acidic environment generated by CO₂ not only destroyed the crosslinking points within the gel network but also weakened the intermolecular forces between polymer chains, leading to a more relaxed gel structure and a reduction in viscoelastic modulus. In contrast, as shown in Fig. 5(b), after CO₂ aging, the elastic modulus of the CO₂-resistant nano-reinforced gel increases from 16.04 Pa to 24.99 Pa, while the viscous modulus rises from 2.33 Pa to 3.22 Pa. This behavior is similar to that observed with increased crosslinker concentration, indicating that the crosslinking density of the CO₂-resistant nano-reinforced gel increases after aging in the CO₂ environment, resulting in a higher elastic modulus and viscous modulus. The physical

crosslinking network within the CO₂-resistant nano-reinforced gel effectively mitigates the damage caused by CO₂, thereby maintaining the stability of the gel system.

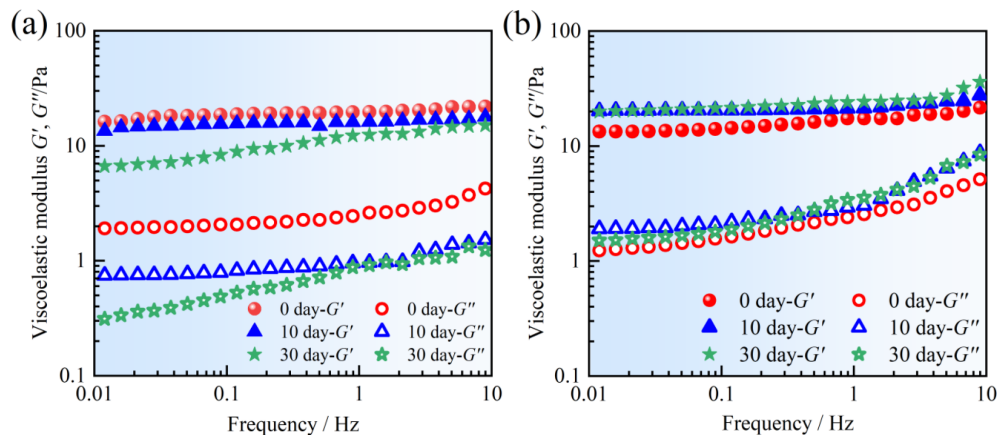


Figure 5—Viscoelastic modulus of different gel systems after CO₂ aging: (a) Conventional HPAM gel; (b) CO₂-resistant nano-reinforced gel.

CO₂ channeling control capability

To evaluate the effectiveness of CO₂-resistant nano-reinforced gel in preventing CO₂ channeling, core flooding experiments were conducted. Table 1. lists the core parameters, and Fig. 6. illustrates the pressure variation during the plugging process of CO₂-resistant nano-reinforced gel in a low-permeability core.

Table 1—The Core parameters.

NO.	The Core parameters				Gel composition	
	Length /mm	Diameter /mm	Permeability / (10 ⁻³ μm ²)	Porosity /%	Polymer concentration / wt%	Crosslinking agent concentration / wt%
Core 1	99.45	25.17	44.97	13.99	0.5	0.6
Core 2	99.13	25.07	41.71	13.27	0.5	0.8

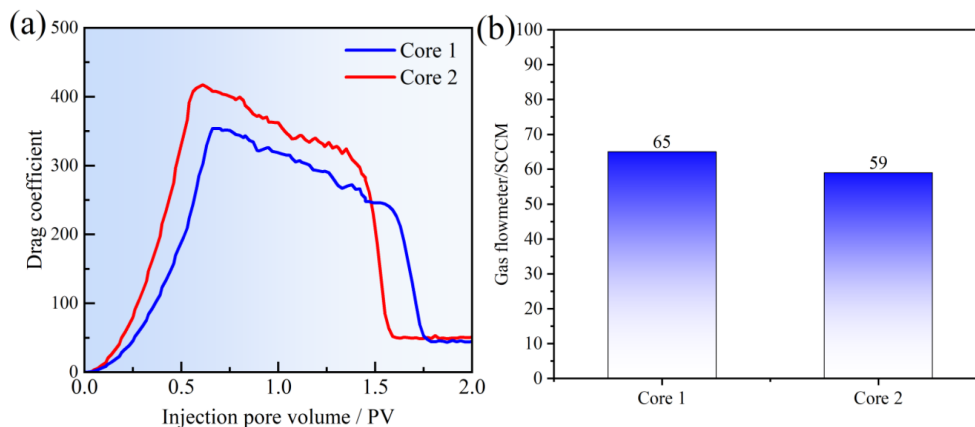


Figure 6—Results of the core flooding experiment: (a) Drag coefficient curve of the core at different injection pore volume (Core 1); (b) Changes of gas flow at the output end after CO₂ injection of 2.0PV.

After injecting 0.5 PV of gel-forming solution, the core was sealed and aged at 90 °C, allowing the solution to gradually transform into a stable solid gel through chemical crosslinking reactions, thereby forming a high-strength plugging barrier. Once the gel was fully crosslinked and solidified, CO₂ was re-

injected into the core. The resulting differential pressure displacement curve, shown in Figure 6, includes four distinct stages of pressure variation: Elastic deformation and pressure rise stage: At the initial stage of CO₂ displacement, the pore throats within the core were already plugged by the CO₂-resistant nano-reinforced gel, significantly hindering CO₂ flow and causing a rapid rise in differential pressure. During this stage, the pressure increased progressively and the pressure curve remained relatively smooth, indicating that under the stress from the displacing CO₂, the CO₂-resistant nano-reinforced gel mainly underwent elastic deformation rather than displacement or failure. This led to cumulative and continuous pressure build-up. Experimental results also showed that CO₂-resistant nano-reinforced gel systems with higher polymer and crosslinker concentrations exhibited faster pressure increases. This is because gels with higher elastic modulus can better maintain their shape under external forces, deforming less and compressing CO₂ into the confined injection space, thus causing a sharp pressure rise. Localized breakthrough and pressure fluctuation stage: Due to the large mobility contrast between the CO₂-resistant nano-reinforced gel-forming solution and the displacing CO₂, some pores received less gel and had relatively weaker plugging strength. These regions were more easily penetrated by CO₂ once breakthrough occurred. As CO₂ injection continued, the CO₂-resistant nano-reinforced gel in these weakly plugged areas was displaced or even breached, leading to fluctuations in the differential pressure curve. Channeling formation and pressure drop stage: With further increases in displacement pressure and declining overall gel strength, more locations of the CO₂-resistant nano-reinforced gel began to shift or break through. CO₂ gradually broke through the gel to form a path of least resistance from the injection end to the production end - a new channeling channel. Stabilized at 59–65 SCCM. At this point, large amounts of gas appeared at the production end, with a sharp increase in flow rate observed on the gas flowmeter. Simultaneously, the differential pressure began to decline rapidly. Stable flow and pressure stabilization stage: As CO₂ injection continued, the differential pressure gradually stabilized after its drop, and gas flow at the production end ceased increasing. Compared to the state before plugging, the newly formed channeling channels still contained residual CO₂-resistant nano-reinforced gel, which reduced the effective cross-sectional area and increased flow resistance. Therefore, even though new channels formed after breakthrough, the final differential pressure remained higher than in the pre-plugging condition.

As shown in Fig. 7., with increasing aging time, the gas channeling development coefficient of the CO₂-resistant nano-reinforced gel after breakthrough first decreases and then stabilizes, eventually reaching a stable value of 0.81. In contrast, the gas channeling development coefficient of the HPAM polymer gel significantly increases after breakthrough, reaching 0.93. This difference is mainly attributed to the change in gel strength of the CO₂-resistant nano-reinforced gel during aging. In a CO₂ environment, the crosslinked network structure of the CO₂-resistant nano-reinforced gel gradually strengthens, resulting in improved mechanical strength. Consequently, even after breakthrough, CO₂ channeling is still effectively suppressed, leading to a decrease in the gas channeling development coefficient. As the aging time increases, the crosslinking network of the CO₂-resistant nano-reinforced gel tends to stabilize, and the further increase in strength becomes marginal, causing the declining trend in gas channeling development coefficient to level off. The CO₂-resistant nano-reinforced gel can form a stable plugging effect under CO₂ exposure, with a breakthrough pressure gradient reaching 40.25 MPa/m. Within a certain aging period, it continues to effectively suppress CO₂ channeling. In contrast, the HPAM polymer gel experiences a decrease in strength after aging, which hampers its ability to suppress channeling following breakthrough.

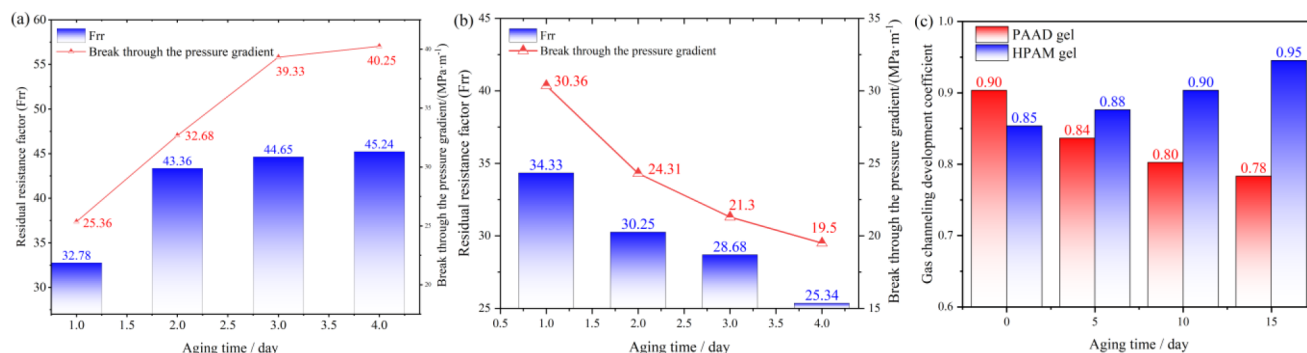


Figure 7—Plugging ability of polymer gel in matrix core after different aging time: (a) Residual resistance factor and break through the pressure gradient (PAAD gel); (b) Residual resistance factor and break through the pressure gradient (HPAM gel); (c) Gas channeling development coefficient.

Granulation of the CO₂-resistant nano-reinforced gel

Fig. 8. presents gel dispersions with different particle sizes prepared from the gel. The gel dispersion can be tailored by adjusting preparation process parameters, specifically by applying mechanical shear and physical rounding to the preformed gel system at different shear rates. This enables the production of gel dispersions with varying particle sizes to meet the demands of different reservoir conditions. The resulting gel dispersion exhibits a particle size distribution ranging from 600 nm to 10 μ m, with the particles uniformly distributed within the liquid phase.

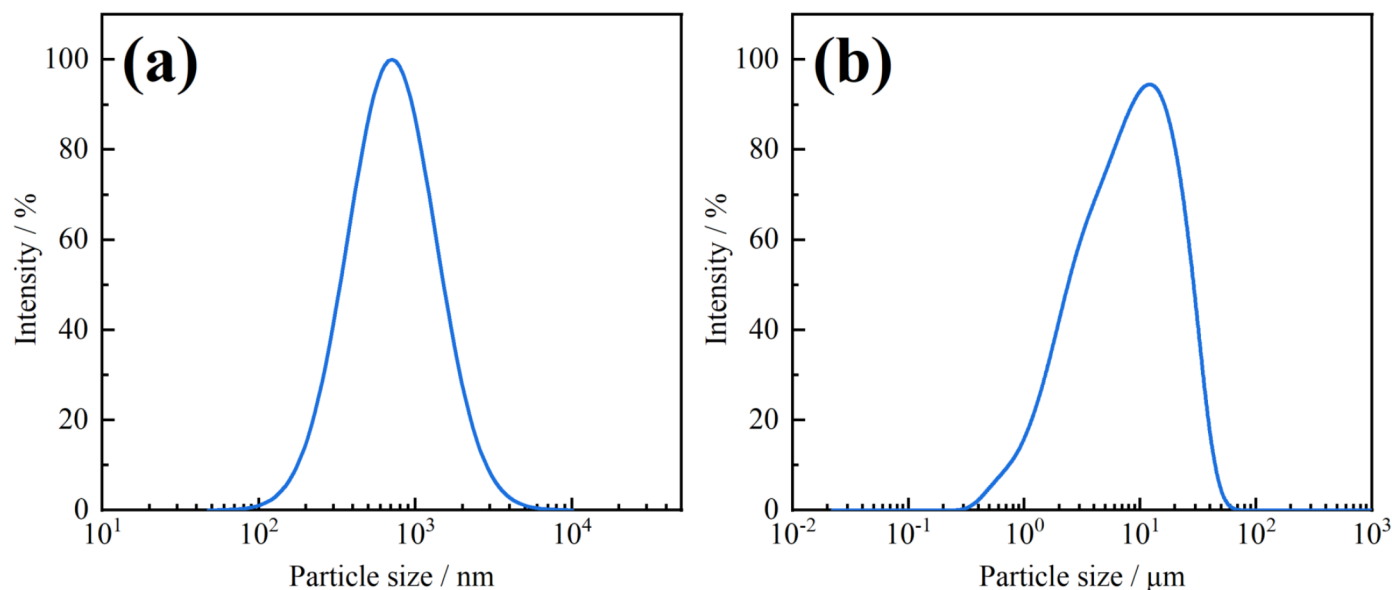


Figure 8—The CO₂-resistant nano-reinforced gel dispersions with different particle sizes: (a) Nanometer scale; (b) Micron order; (c) Millimeter grade.

Evaluation of field experiment

In field applications, effective CO₂ channeling control operations must address not only the high-pressure-difference zones near the wellbore and within fractures, but also ensure control over deep reservoir channeling in more distal areas, in order to achieve enhanced sweep efficiency and improved oil displacement. Therefore, considering the pressure field distribution characteristics within the reservoir and the differing channeling control requirements of near- and far-wellbore regions, a stepwise deep CO₂ channeling control technique has been proposed. This method is based on the developed CO₂-resistant nano-reinforced gel and its particle-forming technique. By combining two types of CO₂ channeling

control systems, the approach achieves dual objectives of "strong near-wellbore plugging and deep far-well regulation," thereby enhancing the overall effectiveness of CO₂ channeling control: (1) The CO₂-resistant nano-reinforced gel: Primarily designed for near-wellbore zones and fracture environments with high pressure differentials. It features high strength and strong resistance to erosion, enabling robust and stable CO₂ sealing performance in heterogeneous and fractured reservoirs. (2) The CO₂-resistant nano-reinforced gel dispersions: Intended for far-wellbore zones and low-permeability reservoir environments. With low viscosity and controllable particle size, it enables effective deep CO₂ channeling control through reservoir-matched optimization.

Fig. 9. shows the injection pressure and displacement curves during the field operation in the oilfield. During the injection process, the pressure steadily increased, reaching 25.4 MPa at the end of injection. This indicates not only that the plugging system possesses good injectivity and mobility, enabling smooth entry into the deep formation, but also that the plugging materials effectively achieved the intended deep-profile control and channeling prevention in the reservoir.

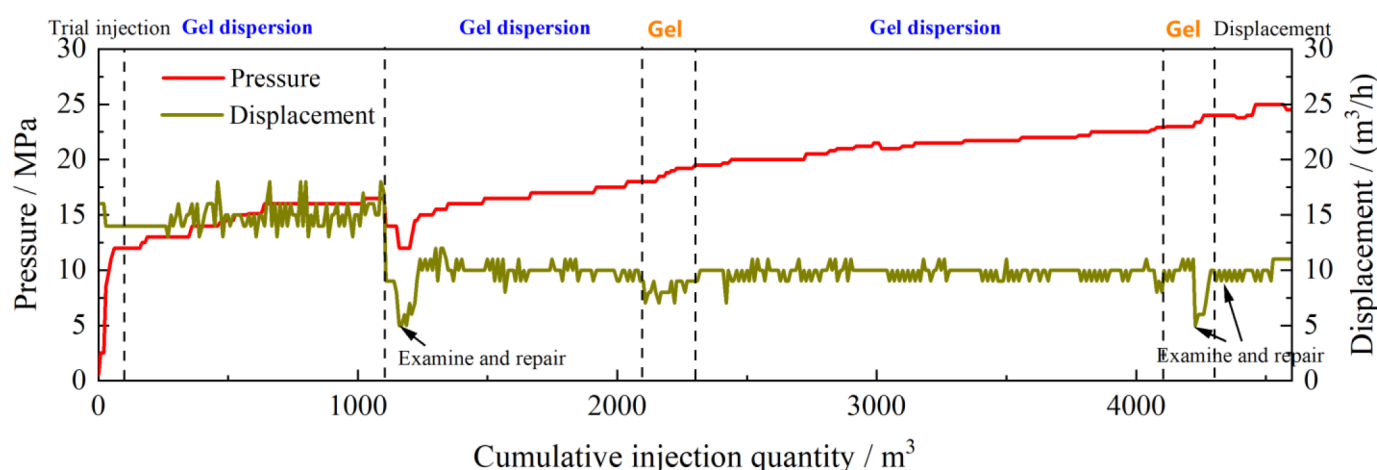


Figure 9—The CO₂-resistant nano-reinforced gel dispersions with different particle sizes: (a) Nanometer scale; (b) Micron order; (c) Millimeter grade.

Conclusions

This study developed a crosslinked polymer gel system for mitigating CO₂ channeling during geological CO₂ utilization, with a focus on its thermal stability, shear resistance, rheological behavior, and CO₂ resistance. The stabilization mechanism of the gel system in CO₂ environments was elucidated, and its channel-blocking performance was validated through both core flooding experiments and field tests. The main conclusions are summarized as follows:

1. The PAAD polymer was synthesized via free-radical polymerization in aqueous solution using acrylamide (AM), 2-acrylamido-2-methylpropanesulfonic acid (AMPS), and dimethylaminoethyl methacrylate (DMAEMA). The PAAD polymer molecules formed supramolecular physical crosslinking structures through electrostatic attraction between sulfonate groups ($R-SO_3^-$) and protonated tertiary amine groups ($R-N^+(CH_3)_2$).
2. Through a systematic investigation of the polymer-to-crosslinker ratio and the viscoelastic properties of the gel, the optimal formulation was determined to be 0.5 wt% PAAD polymer and 0.8 wt% crosslinker, with an optimal HMTA-to-hydroquinone ratio of 1:1. Additionally, 0.2 wt% nano-silica was incorporated to enhance the spatial network structure of the gel.
3. After aging for 30 days in a 90 °C, 15 MPa CO₂ environment, the elastic modulus of the CO₂-resistant nano-reinforced gel reached 24.99 Pa, significantly higher than that of conventional gels (~10.49 Pa).

4. Core flooding experiments showed that the CO₂ breakthrough pressure gradient of the CO₂-resistant nano-reinforced gel reached 40.25 MPa/m, and the residual resistance factor reached 45.24, both significantly higher than those of the HPAM gel, which were 19.5 MPa/m and 25.34, respectively. After CO₂-resistant nano-reinforced gel injection, the gas channeling development coefficient is reduced to 0.78. Furthermore, gel dispersions with particle sizes ranging from 600 nm to 10 μm were prepared using mechanical shearing and physical rounding techniques.
5. Field trials in oilfields confirmed that the gel system, based on a stepwise deep-profile plugging strategy (Strong gel blocking near the wellbore and dispersion-assisted deep regulation), effectively sealed different channeling zones. After injection, the pressure rose to 25.4 MPa, indicating that the plugging materials achieved the desired deep channel-blocking effect within the reservoir.

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